ACADEMIC PROJECT REPORT · 2026

Uttar Pradesh E-Waste Management

Data Validation, Regulatory Context, and Analytical Scorecard

20.5/100	106	15/15	7
Project Score	CPCB Registered Recyclers	Regulatory Compliance	Assessment Categories

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Live Dashboard: ewaste-management-dashboard.vercel.app
Dataset: State_EWaste_Tracker_Final.xlsx
Source Code: github.com/kasim672/ewaste-dashboard

CHAPTER 1

Abstract

This report evaluates selected aspects of e-waste management in Uttar Pradesh using the project workbook State_EWaste_Tracker_Final together with publicly available regulatory evidence. The analysis covers generation and inventory, collection, formal recycling infrastructure, regulatory compliance, informal-sector integration, environmental safeguards, and awareness.

A forensic review of the workbook found that the project score of **20.5/100** is reproducible from its seven weighted categories, but the scoring framework itself is a **study-specific analytical model** rather than an official Central Pollution Control Board (CPCB) state-rating framework. Several figures used in an earlier draft also required correction or qualification.

In particular, the workbook does not contain primary generation data supporting the previously stated 466,578 tonnes of annual e-waste generation, and the resulting 80.78% collection/processing ratio therefore cannot be presented as a verified state indicator.

The workbook records **388,000 tonnes** per annum of recycling capacity and **376,896.74 tonnes** processed, whereas later CPCB material filed before the National Green Tribunal reports **106** registered recyclers in Uttar Pradesh with aggregate capacity of approximately **1,061,557.24 tonnes** per annum.

The study therefore treats workbook-derived values as project data and distinguishes them from externally verified regulatory statistics. The findings emphasize the importance of data provenance, consistent terminology, documented benchmarks, formal-sector monitoring, and transparent treatment of uncertainty in environmental performance assessment.

KEYWORDS

Electronic waste; e-waste management; Uttar Pradesh; CPCB; UPPCB; extended producer responsibility; recycling; environmental governance; data validation

CHAPTER 2

Introduction

Electronic waste consists of discarded electrical and electronic equipment and associated components that have reached the end of their useful life. It presents a dual management challenge: discarded equipment contains recoverable materials of economic value while inappropriate dismantling, burning, chemical extraction, or disposal can create environmental and occupational-health risks.

India's current regulatory framework is principally governed by the **E-Waste (Management) Rules, 2022**, notified by the Ministry of Environment, Forest and Climate Change (MoEFCC) and effective from 1 April 2023. The framework places significant emphasis on Extended Producer Responsibility (EPR), registration of relevant entities, recycling targets, and management through registered recyclers [1], [2].

This project evaluates Uttar Pradesh using an Excel-based analytical scorecard prepared for the State E-Waste Tracker project. The workbook combines operational indicators with a seven-pillar, 100-point scoring model. Because an earlier version of the report described this model as an official assessment framework, the present study re-examines both the underlying dataset and the external evidence before interpreting the score.

The objective is therefore not merely to reproduce the dashboard. It is to determine which conclusions are supported by the project dataset, which require external verification, and which should be treated as assumptions or limitations.

CHAPTER 3

Scope and Objectives

The assessment has four objectives:

- Validate the major quantitative values used by the project dashboard and report.
- Distinguish project-derived metrics from official regulatory statistics.
- Evaluate the internal 100-point scorecard transparently without presenting it as an official CPCB rating.
- Identify data, regulatory, infrastructure, environmental, and reporting gaps relevant to e-waste management in Uttar Pradesh.

The analysis is primarily state-level. District or site-specific conclusions are included only where the available dataset or cited evidence supports them.

CHAPTER 4

Regulatory and Institutional Context

The E-Waste (Management) Rules, 2022 replaced the earlier regulatory framework and established an EPR-based system involving producers, manufacturers, refurbishers, and recyclers [1]. CPCB operates the national E-Waste EPR Portal and performs central regulatory functions, while State Pollution Control Boards, including the Uttar Pradesh Pollution Control Board (UPPCB), retain important inspection, enforcement, and state-level implementation responsibilities.

The MoEFCC Annual Report 2024–25 notes that the 2022 Rules have been in force since 1 April 2023 and cover **106** categories of electrical and electronic equipment [2].

CPCB material filed before the National Green Tribunal provides more recent evidence regarding registered recycling infrastructure. The filing reports **106 recyclers** registered on the CPCB E-Waste EPR Portal in Uttar Pradesh and an aggregate state recycling capacity of approximately **1,061,557.24 tonnes per annum** [3].

These official regulatory figures are used in this report as external validation data and are kept separate from the values contained in the project workbook.

CHAPTER 5

Data Sources

4.1 Project Dataset

The primary analytical dataset is:

- *State_EWaste_Tracker_Final(1).xlsx* — online copy: View dataset

The workbook contains eight principal sheets: a scorecard, instructions, and seven indicator-specific worksheets covering generation and inventory, collection, recycling infrastructure, regulatory compliance, informal-sector integration, environmental and health safeguards, and awareness/outreach.

The accompanying analysis is also published as an interactive dashboard and as versioned source code, listed below for reference:

- **Live Dashboard:** ewaste-management-dashboard.vercel.app
- **Source Code:** github.com/kasim672/ewaste-dashboard

4.2 External Evidence

External validation prioritizes primary or authoritative sources:

- Ministry of Environment, Forest and Climate Change
- Central Pollution Control Board
- Uttar Pradesh Pollution Control Board, where documentary evidence is available
- National Green Tribunal records containing CPCB/UPPCB submissions

Secondary material is used only for contextual environmental discussion where primary state-specific measurements are unavailable.

CHAPTER 6

Methodology

5.1 Workbook Validation

Major values appearing in the original report were traced back to the workbook. Values were classified as:

- **Workbook-supported:** directly present or reproducibly calculated from workbook data
- **Externally verified:** supported by authoritative regulatory evidence
- **Derived but conditional:** arithmetically correct only if an unsupported input is assumed
- **Unsupported:** neither established by the workbook nor adequately supported by an external source

This distinction prevents a calculated value from appearing more certain than its inputs.

5.2 Terminology

The following terms are treated separately:

- **Generation** refers to the quantity of e-waste arising within a defined population and period.
- **Collection** refers to e-waste entering a documented collection or channelization system.
- **Processing/recycling** refers to quantities actually treated by authorized or registered facilities.
- **Installed or authorized capacity** represents potential annual processing capability and must not be treated as actual recycling output.

These distinctions are important because the earlier project report used “collected” and “processed” too interchangeably.

5.3 Project Scorecard

The workbook allocates 100 points across seven categories. The total is: $S = \Sigma si$, where si is the score awarded to each indicator. Therefore,

$$S = 0 + 0 + 0 + 15 + 0 + 5 + 0.5 = 20.5$$

The arithmetic is reproducible. However, no authoritative CPCB or MoEFCC source identified during verification establishes this exact seven-category weighting as an official state assessment system. The 20.5/100 result must therefore be interpreted as a project-specific analytical score.

CHAPTER 7

Data Validation and Provenance

Table below summarizes the status of the major metrics used in the project.

#	Category	Weight	Score Earned	Achievement
1	Generation & Inventory	15	0	0%
2	Collection Performance	20	0	0%
3	Formal Recycling & Treatment	20	0	0%
4	Regulatory Compliance	15	15	100%
5	Informal Sector Integration	10	0	0%
6	Environmental & Health Safeguards	10	5	50%
7	Awareness & Outreach	10	0.5	5%
TOTAL SCORE		**100**	**20.5**	**20.5%**

The validation exercise shows why data provenance matters. A number can be internally consistent without being externally established.

CHAPTER 8

Results

7.1 Generation and Inventory

The most significant data limitation concerns e-waste generation. The project workbook's generation/inventory sheet does not contain sufficient populated category-wise data to reproduce the previously reported total of 466,578 tonnes.

Consequently, this report does not treat 466,578 tonnes as a verified estimate of annual Uttar Pradesh e-waste generation. It may be retained in the project only if its original source, reference year, estimation method, and scope are documented. This limitation also affects any rate using generation as the denominator.

7.2 Collection and Processing

The workbook records 376,896.74 tonnes as the relevant collection quantity and also records 376,896.74 tonnes as processed in the recycling-infrastructure worksheet.

The arithmetic behind the earlier report's calculation is correct. The interpretation is not independently validated because the denominator is unsupported by the workbook. Accordingly, 80.78% should be described as a conditional project-derived ratio, not an established collection efficiency for Uttar Pradesh.

The equality of the workbook's collected and processed quantities should also be treated cautiously. It may indicate that the same aggregate value was reused across worksheets rather than independently measured collection and processing stages.

7.3 Recycling Infrastructure

The project workbook reports a recycling capacity of 388,000 tonnes/year and a processed quantity of 376,896.74 tonnes. Comparing those two workbook values gives an implied utilization of approximately 97.1%.

This is a valid calculation for the project dataset, but it should not be interpreted as utilization of all registered recycling capacity in Uttar Pradesh.

CPCB's later regulatory evidence reports **106 registered recyclers** with aggregate capacity of approximately **1,061,557.24 tonnes per annum** in Uttar Pradesh [3]. This is approximately **2.74 times** the capacity represented by the workbook.

The difference indicates that the workbook is incomplete, based on a different reference period, based on a selected subset of facilities, or constructed from illustrative/project data. Without additional provenance, it cannot represent the current complete state recycling inventory.

7.4 Regulatory Compliance

The workbook awards the full **15/15** points under regulatory compliance.

One input associated with this category is the figure of 9,956 registered producers. CPCB evidence indicates that this figure refers to producers registered on the national EPR portal rather than producers located specifically in Uttar Pradesh [3]. It should therefore not be used as evidence of state-specific producer registration.

CPCB material filed before the NGT nevertheless provides evidence of active regulatory oversight in Uttar Pradesh. UPPCB reported six non-complying recycling units during physical inspections and issued show-cause notices to those units [4].

This demonstrates enforcement activity, but it does not by itself establish complete regulatory compliance across the state. Consequently, the project's 15/15 score should be interpreted as the outcome of its internal scoring criteria rather than a finding that Uttar Pradesh has achieved complete regulatory compliance.

7.5 Informal-Sector Integration

The workbook assigns **0/10** to informal-sector integration. Informal e-waste processing is an important environmental and governance concern in India, but the project dataset does not contain sufficient quantitative evidence to measure the size of the informal sector in Uttar Pradesh or establish a defensible statewide integration rate.

The earlier report identified specific clusters and attributed particular practices to them. Such site-specific statements should be retained only where they can be linked to a documented government action, court record, field survey, or peer-reviewed study.

The zero score can therefore be reported as a project score arising from missing or unmet documented criteria, rather than proof that no integration initiatives exist anywhere in Uttar Pradesh.

7.6 Environmental and Health Safeguards

The workbook awards **5/10** for environmental and health safeguards.

E-waste may contain or release hazardous substances including lead, cadmium, chromium compounds, mercury, brominated flame retardants, and other contaminants when improperly handled. These risks justify environmental monitoring and safe processing requirements.

However, general evidence that e-waste can release hazardous substances does not establish contamination at a particular Uttar Pradesh site. The earlier report's statements that specified locations showed elevated Pb, Cd, and Cr concentrations are therefore not reproduced as factual findings because the project dataset contains no sampling results and no sufficiently specific source was established during verification.

The corrected conclusion is narrower: informal or uncontrolled processing creates a credible environmental risk, but this project does not contain primary environmental sampling data sufficient to quantify local contamination.

7.7 Awareness and Outreach

The workbook awards **0.5/10** for awareness and outreach. This very low value should be understood as a score under the project's own documentation/benchmark system. The dataset does not provide a comprehensive statewide inventory of awareness activities sufficient to conclude that public awareness itself is only five percent of an objective optimum.

CHAPTER 9

Discussion

The principal finding is not simply that Uttar Pradesh obtains a low score. It is that the quality and provenance of the available data strongly influence the score.

The 20.5/100 project result is dominated by three categories: regulatory compliance contributes **15** points, environmental safeguards contribute **5** points, and awareness contributes **0.5** points. Four categories receive zero. In several cases, however, zero reflects missing documentation or an unmet project benchmark rather than direct evidence of zero operational performance.

This creates an important distinction between performance failure and evidence failure.

For example, the project workbook records substantial recycling activity while simultaneously awarding zero points to formal recycling and treatment. Without a clearly documented scoring rule, a reader may reasonably view those findings as contradictory. The scorecard should therefore state explicitly whether a zero means:

- No activity occurred.
- A benchmark was not reached.
- Evidence was incomplete.
- The required documentation was absent.

The comparison with CPCB data further demonstrates that the workbook should not be treated as a complete official state inventory. The official figure of **106 recyclers** and approximately **1.062 million tonnes/year** of capacity is substantially larger than the workbook's 388,000 tonnes/year [3].

The scorecard remains useful as an educational analytical model, provided its limitations are explicit.

CHAPTER 10

Assumptions and Limitations

This study has several limitations.

- First, the workbook does not contain sufficient populated generation data to reproduce the earlier 466,578-tonne estimate. Ratios dependent on that value are therefore conditional.
- Second, the workbook mixes raw observations, aggregate values, benchmarks, and scores. Some values cannot be independently reconstructed from lower-level records.
- Third, the 100-point weighting system is project-developed. The assessment found no basis for describing the exact seven-pillar model as an official CPCB state scorecard.
- Fourth, workbook and regulatory values may refer to different reporting periods. Regulatory registration and capacity data are dynamic and should always be associated with a reference date.
- Fifth, the study contains no primary environmental sampling. It cannot independently establish concentrations of Pb, Cd, Cr, or other contaminants at specific sites.
- Sixth, absence of evidence in the workbook does not necessarily indicate absence of activity in the state. This is particularly important for informal-sector integration, awareness campaigns, and district-level collection systems.

CHAPTER 11

Recommendations

10.1 Improve Data Provenance

Every major workbook input should contain a source, reference year, unit, and retrieval date. External values should not be entered as unexplained constants. A dedicated Data_Sources worksheet should record:

- Metric name
- Value
- Unit
- Geographic scope
- Reporting year
- Source organization
- Document or portal reference
- Verification status

10.2 Separate Raw Data from Calculated Indicators

Raw observations should be stored separately from formulas and scorecard outputs. This would make it possible to reproduce every dashboard indicator and identify whether a problem originates in data entry, calculation, or interpretation.

10.3 Document Scorecard Benchmarks

If the 100-point scorecard is retained, each category should specify: indicator definition, formula, maximum points, benchmark, benchmark source or rationale, missing-data treatment, and scoring rule. This converts the scorecard from an opaque rating into a reproducible analytical model.

10.4 Use Precise Waste-Flow Terminology

Generation, collection, channelization, processing, recycling, and capacity should not be used interchangeably. Each represents a different stage of the material flow.

10.5 Reconcile Recycling Infrastructure Data

The workbook's 388,000-tonne capacity should be reconciled with CPCB's later figure of approximately **1,061,557.24 tonnes/year** for 106 registered Uttar Pradesh recyclers [3]. If the workbook represents only selected facilities, that scope should be stated explicitly.

10.6 Strengthen Regulatory Indicators

National EPR registration totals should not be used as Uttar Pradesh-specific indicators. State-level compliance should instead rely on appropriate measures such as registered facilities, inspection outcomes, notices, environmental compensation, compliance status, or other documented UPPCB/CPCB actions.

10.7 Avoid Unsupported Environmental Claims

Site-specific contamination or health statements should require site-specific evidence. Where such evidence is unavailable, the report should describe the environmental hazard as a risk rather than an observed local condition.

10.8 Preserve Uncertainty

A research report should distinguish measured values, estimates, assumptions, and unknowns. Explicit uncertainty improves credibility more than filling gaps with apparently precise numbers.

CHAPTER 12

Conclusion

The State E-Waste Tracker project provides a useful framework for combining environmental indicators into a single dashboard and scorecard, but its results require careful interpretation.

The workbook reproducibly yields a project score of **20.5/100**. That score is valid as the output of the project's own seven-pillar model, but the framework should not be represented as an official CPCB state rating.

The forensic review identified several important corrections. The workbook does not independently support the previously reported 466,578 tonnes of annual generation, making the associated 80.78% ratio conditional. The 9,956 producer figure is national rather than Uttar Pradesh-specific. The workbook's 388,000 tonnes/year recycling capacity also differs substantially from CPCB evidence reporting **106 registered recyclers** and approximately **1,061,557.24 tonnes/year** of aggregate capacity in Uttar Pradesh.

The strongest conclusion of the study is therefore methodological. Reliable e-waste assessment depends not only on collecting numbers but on documenting their provenance, defining each material-flow stage precisely, separating project assumptions from regulatory facts, and making scoring rules reproducible.

With these corrections, the dashboard and workbook can be presented as a transparent academic analytical exercise rather than as an unsupported official state assessment. The same approach also provides a stronger foundation for future versions of the dataset as more complete generation, collection, recycling, enforcement, and environmental data become available.

CHAPTER 13

References

[1] Ministry of Environment, Forest and Climate Change, Government of India, E-Waste (Management) Rules, 2022, Gazette of India, 2 November 2022.

[2] Ministry of Environment, Forest and Climate Change, Government of India, Annual Report 2024–25, section on hazardous substances and e-waste management.

[3] Central Pollution Control Board, Action Taken Report on behalf of CPCB in EA No. 04 of 2024 in OA No. 512 of 2018, Shailesh Singh v. State of Uttar Pradesh and Others, filed before the National Green Tribunal. The filing reports 106 registered recyclers in Uttar Pradesh and aggregate capacity of approximately 1,061,557.24 TPA.

[4] Central Pollution Control Board, Report in EA No. 04 of 2024 in OA No. 512 of 2018, Shailesh Singh v. Government of Uttar Pradesh and Others, filed before the National Green Tribunal. The report records UPPCB show-cause notices issued to six non-complying recycling units.

[5] Central Pollution Control Board, E-Waste EPR Portal and implementation materials, Government of India.

[6] State E-Waste Tracker Project, State_EWaste_Tracker_Final(1).xlsx, project dataset, accessed July 2026.

CHAPTER 14

Appendix A: Project Score Interpretation

The project's score of 20.5/100 should be cited as:

CORRECT CITATION

“Uttar Pradesh obtained 20.5/100 under the analytical scorecard developed for this study.”

It should not be cited as:

INCORRECT CITATION

“CPCB officially rates Uttar Pradesh 20.5/100.”

The first statement describes the model accurately. The second attributes the model to a regulator without evidence.

CHAPTER 15

Appendix B: Key Reproducible Calculations

Conditional collection/processing ratio using the unverified generation estimate:

$$R = (376{,}896.74 / 466{,}578) \times 100 = 80.78\%$$

Workbook capacity utilization:

$$U = (376{,}896.74 / 388{,}000) \times 100 = 97.14\%$$

Project score:

$$S = 0 + 0 + 0 + 15 + 0 + 5 + 0.5 = 20.5$$

These calculations are mathematically reproducible; their substantive interpretation remains dependent on the provenance and definitions of the underlying inputs.